

AUTOMATIC RAILWAY GATE CONTROL SYSTEM USING IR SENSOR

PHASE 5 — TESTING & VALIDATION REVIEW (TVR)

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1. Phase 5 Objective

Phase 5 evaluates the Version 1 prototype developed in Phase 4. The testing stage focuses on functional operation, performance evaluation, test cases, error analysis, debugging, power consumption, reliability and future improvements. All numerical observations must be entered from actual measurements obtained from the team's prototype.

2. Prototype Under Test

The system under test consists of the dual-IR sensing arrangement, Arduino Uno controller, SG90 servo-driven barrier, LED warning indicator, piezo buzzer and common 5 V/GND supply arrangement established in Phase 3 and implemented in Phase 4.

- IR Sensor 1 (D2): approach-side detection input.
- IR Sensor 2 (D3): departure-side detection input.
- Servo (D9): barrier actuator.
- LED (D7 through 220 Ω resistor): visual warning.
- Buzzer (D8): audible warning.
- Arduino USB/Serial Monitor: development and event-log path.

3. Functional Testing Plan

The following test cases are prepared for execution on the actual prototype. The Pass/Fail column should be completed only after performing each test.

Test ID	Test condition	Expected result	Actual result	Status
TC-01	Power ON / system initialization	Controller and connected modules initialize correctly.	[Record]	[Pass/Fail]
TC-02	No train/object at approach sensor	System remains in normal monitoring state.	[Record]	[Pass/Fail]
TC-03	Valid approach detection at IR1	Warning is activated and gate-control sequence starts.	[Record]	[Pass/Fail]
TC-04	Short/unwanted sensor interruption	Debounce/validation prevents an invalid trigger.	[Record]	[Pass/Fail]
TC-05	Train/object occupies crossing	Gate remains in the required closed/occupied state according to control logic.	[Record]	[Pass/Fail]
TC-06	Valid departure detection at IR2	Departure is recognized and reopening sequence starts.	[Record]	[Pass/Fail]
TC-07	LED operation	LED responds to the programmed warning/status condition.	[Record]	[Pass/Fail]
TC-08	Buzzer operation	Buzzer responds to the programmed warning/status condition.	[Record]	[Pass/Fail]
TC-09	Servo movement	Barrier moves to the commanded position without mechanical obstruction.	[Record]	[Pass/Fail]
TC-10	Repeated approach/departure cycle	System returns to monitoring state after a complete cycle.	[Record]	[Pass/Fail]

4. Performance Evaluation

Performance evaluation should be based on measurements made during the prototype demonstration. Record the measured values below rather than estimated or assumed values.

Parameter	Measurement to record
Sensor detection response	[Actual measured response time]
Gate closing time	[Actual measured time]
Gate opening time	[Actual measured time]
Debounce/validation threshold	150 ms configured in the Phase 3 design; verify actual firmware value.
Alert lead time	250 ms configured in the Phase 3 design; verify actual firmware value.
Supply voltage	[Measured voltage]
Current consumption	[Measured current]
Successful cycles / total cycles	[Actual count]

5. Error Analysis

Possible prototype-level error sources identified from the earlier design and risk analysis are listed below. The team should mark the observed condition only when it occurs during actual testing.

- IR sensor misalignment or interruption of the intended line of sight.
- False triggering caused by brief interruptions, debris, animals or ambient IR interference.
- Servo movement affected by mechanical mounting or insufficient supply capability.
- Unstable operation caused by loose wiring or an unsuitable power connection.
- Timing mismatch caused by differences between the prototype's configured parameters and the test condition.
- Gate reopening not occurring as expected after departure.

6. Debugging Report

Use this table to document actual faults found during testing and the corrective action taken.

Issue ID	Observed problem	Possible cause	Corrective action	Result
D-01	[Record actual issue]	[Cause]	[Action]	[Verified result]
D-02	[Record actual issue]	[Cause]	[Action]	[Verified result]
D-03	[Record actual issue]	[Cause]	[Action]	[Verified result]

7. Power Consumption Analysis

Power consumption must be measured on the actual Version 1 prototype. Record the supply voltage and current for the controller in normal operation and during the highest observed load condition. The Phase 3 design specifies USB 5 V for bring-up or a 4×AA battery pack for an untethered demonstration.

Measured supply voltage: [____ V] **Measured current:** [____ A]

Peak/maximum measured current: [____ A] **Measurement instrument:** [____]

8. Reliability Testing

Reliability testing should use repeated complete operating cycles. Each cycle should include approach detection, gate operation, crossing occupancy as applicable, departure detection and return to the normal monitoring state.

Trial	Cycle result	Notes
1	[Pass/Fail]	[Actual observation]
2	[Pass/Fail]	[Actual observation]

Trial	Cycle result	Notes
3	[Pass/Fail]	[Actual observation]
4	[Pass/Fail]	[Actual observation]
5	[Pass/Fail]	[Actual observation]
Total	[Successful / total]	[Actual summary]

9. Validation Checklist

- Both IR sensor inputs respond correctly when the sensing path is interrupted.
- Approach detection produces the programmed warning and gate response.
- Departure detection produces the programmed reopening response.
- Servo movement is mechanically smooth and repeatable.
- LED and buzzer provide the intended indication.
- Debounce/validation logic is checked using actual test conditions.
- All actual observations are recorded for the final report.
- Prototype wiring and power connections remain safe and secure during testing.

10. Future Improvements

- Improve sensor mounting and alignment for more consistent beam detection.
- Refine debounce and timing parameters using measured test observations.
- Consider a stronger gate actuator and limit switches for a larger prototype.
- Consider redundant sensing for improved fault tolerance.
- Add backup power and additional fault monitoring for future field-oriented designs.
- Evaluate multi-track, bidirectional and remote-monitoring extensions only as future work, outside the base prototype scope.

11. Testing Evidence to Attach

- Photographs of each important test setup.
- Short video clips or screenshots showing sensor detection and gate movement, where required.
- Serial Monitor/event-log screenshots if used.
- Measured voltage/current records.
- Completed test-case table.
- Completed debugging table.
- Reliability-cycle observations.

12. Phase 5 Conclusion

Phase 5 provides the structured validation stage for the Automatic Railway Gate Control System Using IR Sensor. The Version 1 prototype is evaluated against functional and performance requirements using defined test cases. Actual measured results, faults, corrective actions, power measurements and repeated-cycle observations should be entered before submission. The validated evidence will form the basis for the Final Phase documentation, presentation, demonstration video and live prototype preparation.